

Tharun V. Puthanveettil

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Human-Inspired Robot Learning | Multimodal Learning | Human-Robot Interaction

Research Focus

I want to build robots that understand the people they work with and keep learning from them over time. My research takes inspiration from how people learn and understand one another: a robot should recognize what it does not yet know, learn it from the right person through demonstration, correction, or instruction, and keep what matters. I approach this through multimodal understanding of people, learning from human intervention, explainability of learned policies, and representation and predictive models, evaluated on physical robots.

Research interests: human-inspired robot learning; multimodal understanding of people and human-robot interaction; learning from human demonstration, correction, and instruction; continual and interactive learning; explainability and representation learning for robot policies; vision-language-action policies; world and inverse-dynamics models; compliant manipulation; real-to-sim-to-real transfer; robot-policy evaluation.

Research Threads

Understanding People

PRIOR WORK · 2021–2024

How can a robot recognize what people are doing and what they mean from incomplete, multimodal evidence?

- Multi-view action recognition and fall detection from human pose under occlusion with GCN–transformer models; under the reported occlusion protocol, multi-view F1 reached 0.82 versus 0.76 for single view (graduate course project). [\[POSEFUSION\]](#)
- [Track Anything Rapter](#): a person specifies a target by click, box, image, or text; foundation-model perception and visual servoing close the loop on a drone.
- Exploratory policy combining demonstration, speech, and gesture for robot instruction on the [NatSGD dataset provided by Sneesh Shrestha \(UMD\)](#); task success and the benefit of gesture remain unvalidated.

Learning from People

SYSTEM IMPLEMENTED; EVALUATION MANUAL

How can robots improve from people's demonstrations and corrections without discarding prior capability?

- Connects takeover, correction capture, data review, conversion, retraining, testing, and controlled rollout through common data and policy interfaces; automated policy evaluation is not yet complete. [\[OVERVIEW\]](#)

Egocentric Experience and Multimodal Interaction

INTERFACE IN DEVELOPMENT

How can embodied human experience be converted into useful robot learning signals?

- Developing a portable egocentric demonstration interface that reuses the synchronization, review, and dataset-conversion infrastructure of the existing robot-learning platform. [\[OVERVIEW\]](#)

Policy Explainability and Representation

EXPERIMENTS IN PROGRESS

What has a policy learned, and which observations, prompts, states, and priors shape its actions?

- Uses attention maps, Grad-CAM, state-vector ablations, prompt comparisons, and pretraining/self-supervised representation comparisons as behavioral diagnostics rather than causal explanations. [\[OVERVIEW\]](#)

Predictive Models, Inverse Dynamics, and Transfer

OFFLINE EVALUATION COMPLETED

How can predicted outcomes and inverse dynamics support action learning on physical robots?

- Adapted a published video-action architecture by retraining its action head on teleoperation data and completed offline evaluation. Physical validation and simulator-trained inverse dynamics remain future work. [\[OVERVIEW\]](#)

Education

Master of Engineering in Robotics, GPA: 3.9/4.0

University of Maryland, College Park

Aug. 2022 - May 2024

Maryland, USA

Selected coursework: Advanced Techniques in Visual Learning and Recognition; Perception for Autonomous Robots; Planning for Autonomous Robots; Control of Robotic Systems.

B.Tech. in Electronics and Communication Engineering, GPA: 8.59/10.0

Vellore Institute of Technology

Jul. 2014 - Apr. 2018

India

Research and Professional Experience

Robotics Engineer – Robot Learning and Deformable Manipulation

Jun. 2024 - Present

General Motors R&D

Michigan, USA

- Architected and implemented a [modular ROS 2 robot-learning platform](#) connecting bimanual teleoperation, multimodal perception, synchronized data capture, simulation, dataset conversion, policy training, manual evaluation, staged testing, and physical deployment.
- Led the human-correction workflow (takeover, correction capture, review, retraining, controlled rollout) and the policy explainability and representation studies summarized under Research Threads; designed common data, policy, and deployment interfaces so each stage can be studied independently.
- Built task-conditioned detection, segmentation, tracking, and pose-servoing services with foundation models; reduced perception latency by approximately 35% and sustained 30–40 FPS through embedding reuse and temporal mask propagation.
- Developed force-aware bimanual manipulation of deformable linear objects using impedance control and force-torque feedback; quantified a twofold increase in failures without elastic-tension regulation.
- Created a digital twin and behavior-tree-driven synthetic-data workflow to characterize policy degradation under sensing, timing, dynamics, actuation, contact, and visual shift.
- Architected the Cartesian motion-policy, safety, and evidence pipeline; supervised its extension to dual-arm collection and rollout with admittance-based compliance, synchronized RGB–wrench data contracts, matched collection/deployment control, and measured-state reseeding.
- Evaluated action-chunking and open-source vision-language-action policies while developing measures of partial progress, failure type, recovery, repeatability, and compliance sensitivity.

Researcher - AI and Robotics

Feb. 2023 - Mar. 2024

CATT Laboratory, University of Maryland

Maryland, USA

- Served as core architect of a [multimodal sequential anomaly-detection pipeline](#) using modality-specific LiDAR, odometry, and network encoders with reconstruction-based scoring. The published system reported 98% accuracy versus 72% for LiDAR alone, and 97% with half the training data.
- Co-developed [Track Anything Rapter](#), a [ROS 2 aerial system](#) accepting click, image, text, and box target specifications through foundation-model detection/segmentation, visual servoing, and PX4 control.

Researcher - Collaborative Robotics and Computer Vision

Jun. 2021 - Jan. 2023

Indian Institute of Science and University of Maryland

India and USA

- Developed a [vision-guided precision-weeding system](#) combining crop/weed detection, root and plant-spread estimation, workspace adaptation, hybrid inverse kinematics, [closed-loop control](#), and [end-effector design](#).
- Developed synthetic-data and augmentation pipelines for weed localization; improved robustness by 20–30% through hard-negative sampling and structured distractors. Co-designed a dual-purpose end effector and adaptive position/torque-control methods for delicate plant geometry.
- Implemented an [exploratory policy](#) combining demonstration, speech, and gesture for robot instruction.
- Developed and evaluated a [U-Net monocular depth model](#) in an IROS 2021 challenge setting.

Project Engineer – AI Innovation

Jul. 2018 - May 2021

CTO Office, Wipro Digital

Bangalore, India

- Served as AI project lead for applied-research prototypes in computer vision, HCI, recommendation, generative modeling, and mobile inference; coordinated quarterly prototype selection and delivery.
- Built tool tracking, mobile vehicle recognition, photogrammetry, [human-pose estimation](#), [virtual try-on](#), and [recommendation](#) components for industrial and retail workflows.

Selected Earlier Technical Projects

Monocular Depth Estimation | *PyTorch, U-Net, transfer learning*

2021

- Built a monocular depth-estimation pipeline and evaluated it in an IROS challenge setting. [\[CODE\]](#)

Implicit Neural Representation and SLIC Segmentation | *PyTorch, TensorFlow, OpenCV*

2023

- Implemented [coordinate-conditioned image outpainting](#) and [deep-feature superpixel segmentation](#), including controlled comparisons with simpler baselines.

Maze Runner and Autonomous Retrieval Robot | *ROS 2, Nav2, visual servoing*

2023

- Built [visual-cue-based maze navigation](#) in *Gazebo* and an [onboard-perception retrieval robot](#) using sensor fusion and visual servoing.

Human-Centered Vision and Interactive AI | *pose, WebVR, recommendation, generative vision*

2019 - 2021

- Developed an [AI cricket coach](#), [virtual try-on system](#), and [personalized style assistant](#) combining pose estimation, action recognition, immersive feedback, recommendation, and generative modeling.

Reinforcement-Learning Cricket Simulation | *adaptive agents and simulation*

2020 - 2024

- Built a simulated bowler agent that adapts to batting styles and field configurations. [\[CODE\]](#)

Publications

(Also published as V. P. Tharun.)

1. A. Zoukarni, M. Noorani, J. S. Baras, J. Mirenzi, **T. V. Puthanveettil**, and C. D. Grody. “Efficient Integration of Domain Knowledge and Data-Driven Methods for Securing Autonomous Cyber-Physical Systems.” IEEE ROSE 2026. DOI: [10.1109/ROSE69181.2026.11570974](https://doi.org/10.1109/ROSE69181.2026.11570974).
2. **V. P. Tharun** and A. R. Chowdhury. “Vision-Based Hybrid IK Task Planning with Feedforward Neural Network for Collaborative Plant-Robot Interaction.” *International Conference on Social Robotics + AI (ICSR + AI 2025)*, LNAI 16132, Springer. DOI: [10.1007/978-981-95-2382-5_18](https://doi.org/10.1007/978-981-95-2382-5_18).
3. M. Noorani, **T. V. Puthanveettil**, A. Zoukarni, J. Mirenzi, C. D. Grody, and J. S. Baras. “Multimodal Anomaly Detection for Autonomous Cyber-Physical Systems Empowering Real-World Evaluation.” *GameSec*, 2024.
4. L. P. Pratap, P. M. Shailendrasingh, A. Anand, and **V. P. Tharun**. “Wall-Climbing Robot Using Soft Robotics.” *ICPCSI*, 2017.

Earlier work in data mining and forecasting (2018–2020): five papers and reports on rainfall forecasting and water-quality assessment. [\[GOOGLE SCHOLAR\]](#)

Preprints and Research Outputs

1. **T. V. Puthanveettil**, A. Singh, Y. Jain, V. Bukka, and S. Arjun S. “Auto-Platoon: Freight by Example.” *arXiv:2405.11659*, 2024. [\[CODE\]](#)
2. **T. V. Puthanveettil** and F. Obaid ur Rahman. “Track Anything Rapter (TAR).” *arXiv:2405.11655*, 2024. [\[CODE + DEMOS\]](#)
3. **Tharun Puthanveettil**, Abhijay Singh, Aniruddh Balram. “Pose Fusion - Multi-View Pose Integration for Comprehensive Action Recognition.” UMD graduate course project. Project poster and preprint. [\[PREPRINT\]](#)
4. **V. P. Tharun** et al. “Application of Mobile Collaborative Robot using Deep Learning in Precision Weed Control.” Manuscript, 2021.
5. **V. P. Tharun** et al. “Event-Based Dynamic Obstacle Avoidance in Outdoor Environments.” IROS 2021 workshop poster.

Technical Skills

Robot learning: PyTorch, TensorFlow, LeRobot, action-chunking policies, vision-language-action models, diffusion/flow policies, imitation learning, self-supervised learning

Robotics and control: ROS 2, ROS, MoveIt 2, impedance/compliance control, visual servoing, behavior trees, bimanual manipulation, force-torque feedback

Simulation and perception: Isaac Sim/Lab, MuJoCo, Gazebo, OpenCV, Open3D, RGB-D cameras, detection, segmentation, tracking, pose estimation

Programming and systems: Python, C++, SQL, Linux, Docker, Git, REST APIs

Intellectual Property, Mentoring, and Communication

Patents: Inventor on three filed patent applications; details withheld pending publication.

Research supervision: Supervised an engineering intern extending the robot-learning platform for compliant bimanual data collection, learning, analysis, and rollout.

FIRST Robotics: Electrical Programming Head Coach, The Irrational Engineers (Jun. 2024–Present); mentor high-school students in controls, computer vision, and software.

Writing: *Human in the Loom*, essays on robot learning, embodied intelligence, and credible research practice. [\[ESSAY ARCHIVE\]](#)